

① Boyle's Law at constant T_{em}

$$V \propto \frac{1}{P} \quad P_1 V_1 = P_2 V_2$$

② ~~Charles~~ Avogadro's Law molar volumes are same at constant P_{ress} and T_{em}

$$V = \frac{V}{n}$$

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

Charles Law at con P_{ress}
 $V \propto T$

③ $PV = nRT \rightarrow$ ideal

$\rightarrow 0.082$ at $V \rightarrow L, P \rightarrow atm$

④ mole fraction and partial pressure

$$X_i = \frac{P_i}{P_{total}}$$

$$P_i = X_i P_{total}$$

$\rightarrow * 1 = \sum X_i$ degrees of freedom

⑤ $K_c = \frac{3}{2} RT$

$J \rightarrow 8.314$

⑥ $\bar{u} = \sqrt{3RT}$

$Xg \leftarrow M \rightarrow$ molecular mass

$$\frac{\bar{u}_b}{\bar{u}_a} = \sqrt{\frac{m_a}{m_b}}$$

⑦ $\left(P_{obs} + \frac{an^2}{V^2} \right) + (V - nb) = nRT \rightarrow$ Real

$$P_{obs} = \left(P - \frac{an^2}{V^2} \right) \quad C.F$$

Chapter 1

$$T_x = T_c + 273.15$$

$$T_c = (T_F - 32^\circ F) \times \frac{5^\circ C}{9^\circ F}$$

$$T_F = (T_c \times \frac{9^\circ F}{5^\circ C}) + 32^\circ F$$

Conversion Factor = $\frac{\text{unit required}}{\text{unit given}}$

Chapter 3

$$\text{Viscosity} = \frac{F \times d}{v \times A} = \boxed{\text{Poise}}$$

$$\eta = \frac{\pi r^4 P \tau}{8 L v}$$

$$\frac{\eta_1}{\eta_2} = \frac{t_1 d_1}{t_2 d_2}$$

$$\eta_{\text{specific}} = \frac{\eta_{\text{H}_2\text{O}}}{\eta_{\text{H}_2\text{O}}}$$

$$P.M = 2RT$$

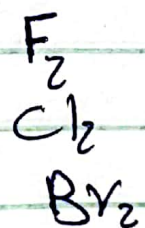
$$P.V = \frac{m}{M} RT$$

★ boiling point ★
★ in London force (non polar) ★

b.p $\propto$ Polarizability

$\propto \uparrow$ size

the stronger



Chapter 4

$$\ln\left(\frac{P_1}{P_2}\right) = \frac{-\Delta H_v}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Note Normal Boiling Point

$$1 \text{ atm} = 760 \text{ torr} = 760 \text{ mmHg}$$

Factors affecting on P_v

1. Molecular mass $\rightarrow$ MM $\uparrow$ $P_v \downarrow$
2. Intermolecular force $\rightarrow$ f $\uparrow$ $P_v \downarrow$
3. Temperature $\rightarrow$ T $\uparrow$ $P_v \uparrow$

Highest + Volatile = Lowest Boiling Point

Chapter 5

$$\Delta E = q + w \quad \text{or} \quad \Delta E = \Delta H + w$$

I. E

Heat

work

Absorbed $\rightarrow$ +ve Into

ΔE

Emitted $\rightarrow$ -ve out

Exothermic $\rightarrow$ -ve Lose

q

Endothermic $\rightarrow$ +ve Gain

on system **or** By Surround $\rightarrow$ +ve

By System **or** on Surround $\rightarrow$ -ve

Chapter 6

Raoult's Law

Solid $\times$ Liquid

$$P_{\text{soln}} = X_{\text{solvent}} \cdot P^{\circ}_{\text{solvent}}$$

Liquid $\times$ Liquid

$$P_{\text{soln}} = (P^{\circ}_A \cdot X_A) + (P^{\circ}_B \cdot X_B)$$

$i = \frac{\text{moles of particles in soln}}{\text{moles of dissolved solute}}$



$$i = 2$$

$$i = 1$$

Glucose

Boiling Points

$$T_b(\text{solvent}) < T_b(\text{solution})$$

$$\Delta T_b = i K_b m_{\text{solute}} = T_b(\text{soln}) - T_b(\text{solvent})$$

Van't Hoff Constant No.

Aqueous soln. $\rightarrow$ Water $T_b = 100^\circ\text{C}$
 $T_f = 0^\circ\text{C}$

Freezing Points

$$T_f(\text{solution}) < T_f(\text{solvent})$$

$$\Delta T_f = -i K_f m_{\text{solute}} = T_f(\text{soln.}) - T_f(\text{solvent})$$

OR

$$\Delta T_f = i K_f m_{\text{solute}} = T_f(\text{solvent}) - T_f(\text{soln.})$$

Elbaram

Osmotic Pressure

$$\pi = M R T i = \frac{n}{V} R T i = \frac{M}{M_w \cdot V} R T i$$

$M \rightarrow$ molarity $R \rightarrow 0.082$ $T \rightarrow$ Kelvin

Concentration

1. Mole fraction (X_i)

$$X_i = \frac{n_i}{n_t}$$

$n_t \rightarrow$ Total

2. mass fraction

$$MF = \frac{w_i}{w_t}$$

3. Molarity (M)

$$M = \frac{n_{\text{Solute}}}{V_{\text{Solution}}} \text{ mol/L, Molar, } M$$

4. Molality (m)

$$m = \frac{n_{\text{Solute}}}{\text{kg } w_{\text{Solvent}}} \text{ mol/kg, molar, } m$$

5. Normality (N)

$$N = \frac{n(\text{eq})_{\text{Solute}}}{V_{\text{Solution}}}$$

$$n(\text{eq}) = n \cdot X$$

no. of ele

H^+

OH^-

$$\text{equivalent weight} = \frac{M_w}{X}$$

Chapter 7

$$K_c = \frac{[P]^n}{[R]^m}$$

$n \rightarrow$ No. of mole of (P)

$$K_p = \frac{[P]^n}{[R]^m}$$

P $\rightarrow$ Product

R $\rightarrow$ Reactant

m no. of mole of (R)

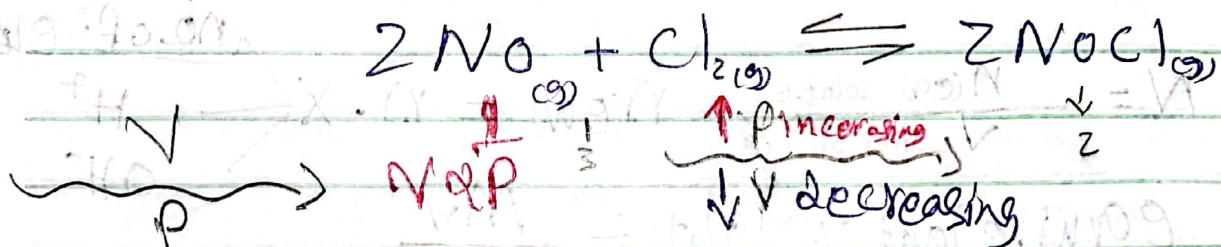
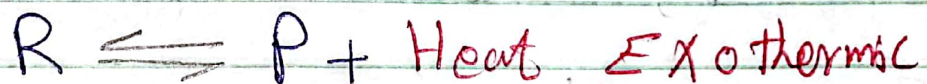
★ The Solid and Liquid No effect on K_c or K_p concentration

K_c or $K_p \rightarrow$ No change except by change Temperature

Relation between (K_c) and (K_p)

$$K_p = K_c (RT)^{\Delta n_{\text{gas}}} \quad \Delta n \rightarrow P - R$$

$$K_c = K_p (RT)^{-\Delta n_{\text{gas}}}$$



شؤون التعليم العالي والبحث العلمي

ElSalam

Chapter 8

→ concentration of acid

$$[H^+]^2 = K_a C_a \quad ; \quad [H^+] = \sqrt{K_a C_a}$$

$$[OH^-]^2 = K_b C_b \quad ; \quad [OH^-] = \sqrt{K_b C_b}$$

$$pH = -\log [H^+] \quad ; \quad pOH = -\log [OH^-]$$

$$pH + pOH = 14$$

$$[H^+][OH^-] = 10^{-14} \rightarrow K_w$$

$$\text{Ionization or dissociation} = \frac{[H^+]}{[C_a]} \text{ OR } \frac{[OH^-]}{[C_b]}$$

* The weak acid have strong conjugate base

$$* K_w = K_a K_b$$

in buffer

$$[H^+] = K_a \frac{[Solute]}{[acid]}$$

$$pH = -\log K_a + \log \frac{[Solute]}{[acid]} \text{ OR } \frac{[con. base]}{[acid]}$$

$$[OH^-] = K_b \frac{[Solute]}{[Base]}$$

$$pOH = pK_b + \log \frac{[Solute]}{[Base]} \text{ OR } \frac{[con. acid]}{[Base]}$$